

Interpreting Differences among Hydraulic Parameters for Different Soil Series by Functional Characterization

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ABSTRACT

Using soil survey data for simulation modeling of water regimes and crop growth requires characteristic expressions for hydraulic characteristics of major soil horizons and soil series. It is unclear at present whether: (i) hydraulic functions based on parameter estimation techniques are adequate for simulations in macroporous soils, and (ii) to what extent different soil horizons and series can be pooled to reduce input data in simulation models. These questions were examined using two methods of expressing measured hydraulic conductivity data in computer simulations with previously validated crop models. Both hydraulic conductivity expressions used parameterized equations derived from one-step outflow data for the -5.0 to -60.0 kPa pressure range. For the 0 to -5.0 kPa range, one method used hydraulic conductivities estimated by extrapolating the parameterized functions, whereas the second method used actual crust test data to account for macropore flow. Differences between the two methods were functionally tested by simulating maize (*Zea mays* L.) yields and fluxes below the root zone in three Costa Rican soil series, using the SWACROP model. Yields were significantly higher when considering structural porosity (Method 2), while fluxes were significantly lower. Thus, extrapolation of one-step outflow data to conditions near saturation seems to be questionable. Using Method 2, maize yields and fluxes below the root zone were compared for three representative soil series to test whether they could be considered as one functional group of soils. Yield data indicated that two rather than one functional group should be distinguished. Flux data were similar for all three series, indicating that only one functional group was necessary.

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Published in Soil Sci. Soc. Am. J. 59:344-351 (1995).

DIGITIZED SOIL MAPS are increasingly being used as input for modeling nutrient movement or crop growth in land areas for which different environmental or agricultural land-use scenarios need to be developed (e.g., Wösten et al., 1985; Breeuwsma et al., 1986; van Lanen et al., 1986). In these maps, soil horizons are characterized in terms of basic physical and chemical parameters needed for simulating crop growth and solute movement. One set of data is usually supplied for each soil type, implicitly assuming that different types of land use do not significantly affect the listed soil properties.

Two approaches are generally taken for using basic soil physical parameters in simulation modeling. The first approach is to use continuous input functions, derived from parameterized equations such as those developed by van Genuchten (1980), with parameters obtained by optimization procedures. Experimental data are usually obtained in a limited range of pressure heads (e.g., -5.0 to -60.0 kPa for the one-step outflow method). Data for wetter and drier conditions are predicted by the parameterized equations (Kool et al., 1985; Parker et al., 1985). The second approach is to use soil physical parameters in a tabular form, allowing a more flexible expression of the physical parameters near saturation, which are difficult to characterize with the parameterized equations mentioned above because of sharp drops in hydraulic conductivity following desaturation of macroporous soils (e.g., Bouma, 1991).

In addition to the question of whether to use tabular or continuous tabular data, it is still unclear whether separate sets of physical data for soil horizons in each

soil series are really needed or if data can be combined for different horizons and soil series. Therefore, a common question to each discipline making contributions to models is the extent to which input parameters can be reduced without significantly affecting the predicted outcome.

These issues are evaluated here within the context of an ongoing project to determine alternative land-use scenarios in the Atlantic Zone of Costa Rica. Crop growth simulation models are being run for seven crops in an area of 50 km² with 75 major soil units. Soil and climate data, as well as information from economic and social studies, provide boundary conditions for various scenarios to be considered (Alfaro et al., 1994).

This study addresses a basic issue encountered by pedologists when applying large quantities of available soil survey data for modern applications. They don't know whether pedological differences are also significant from a functional point of view. To answer this question, they need to test their data by expressing differences in terms of the effect that these differences have on well-defined and practically relevant soil functions. In this study, functionality relates to: (i) simulated water supply and crop growth of maize, and (ii) fluxes below the root zone.

MATERIALS AND METHODS

Soils and Land Use

The research area of the Atlantic Zone Program is concentrated in the northern part of the province of Limón in Costa Rica. The major part of the area is a lowland sedimentation basin, dissected by many streams of variable size. Rainfall increases gradually from the Caribbean coast in the northeast to the volcanic mountain chain in the southwest, but is on average 3000 to 4000 mm y⁻¹. Due to the volcanic origin of the transported and sedimented material, most of the soils in the basin are Andisols or soils with andic properties. Identification, mapping, and classification resulted in the recognition of 75 soil series. In an attempt to reduce the number of soil series being distinguished, parent material, drainage condition, and stage of development have been used to define three major soil groups (Wielemaker and Kroonenberg, 1992): (i) young Holocene soils, well drained, fertile; (ii) young Holocene soils, poorly drained, fertile; and (iii) old Pleistocene soils, well drained, unfertile.

To evaluate the effect of grouping all of the young, fertile, well-drained soil series into one major soil group, three soil series within this group were selected. They span the range of characteristics within the group, particularly in terms of soil texture, which was not considered when distinguishing the three groups. Analytical data for the selected soil series are given in Table 1.

The Ligia soil series (Andic Dystropept) is the oldest soil in the selected group. It is characterized by a brownish black (7.5YR 2/1) to dark brown (10YR 3/3) A horizon with a silty clay loam texture and a strong, very fine subangular blocky, porous structure. The A horizon grades into a brown (10YR 5/3) cambic B horizon with a sandy loam to loamy sand texture and a moderate, very fine to fine, subangular blocky to massive structure. This, in turn, grades into a C horizon consisting of greyish yellow brown (2.5YR 6/3) loose sand. The soil porosity is high to very high in the A horizon, ranging from macro-

Table 1. Analytical data for the soils studied.

	layer	sand	silt	clay	organic matter	bulk density
	m	%				kg m ⁻³
Los Diamantes	0.00–0.05	67	10	23	7.77	†
loamy, mixed, isothermic	0.05–0.20	67	9	24	6.16	810
Vitric Hapludand	0.20–0.65	75	14	11	2.14	910
	0.65–1.05	82	9	9	0.80	1170
	1.05–1.20	94	1	5	0.53	†
Ligia	0.00–0.10	22	47	31	2.14	†
clayey, mixed, isothermic	0.10–0.40	24	31	45	0.80	860
Andic Dystropept	0.40–0.70	40	13	47	0.80	1120
	0.70–0.90	56	5	39	0.80	1120
	0.90–1.20	90	1	9	0.53	†
Montelimar	0.00–0.05	55	25	20	6.16	†
loamy, mixed, isothermic	0.05–0.25	60	22	18	5.77	710
Humic Udivitrant	0.25–0.55	64	25	11	2.14	710
	0.55–0.80	60	27	13	1.65	1160
	0.80–1.05	71	16	13	0.53	†

† Not measured.

to micropores, gradually grading into fine- to medium-sized porosity in the subsoil.

The Los Diamantes soil series (Vitric Hapludand) is characterized by a brownish black (7.5YR 2/1) A horizon, with a loam texture and a moderate, very fine to fine subangular blocky, porous structure, gradually grading into a dull, yellowish brown (10YR 5/6) cambic B horizon with a sandy loam to loamy sand texture and a weak, very fine subangular blocky to massive structure, grading into greyish yellow brown (2.5YR 6/3) granular sand, containing coarse gravel, stones, and boulders. Also, this A horizon contains many fine- to coarse-size pores, gradually grading into few, very-fine- to medium-sized pores in the subsoil. Throughout the soil, slight andic properties can be observed.

The Montelimar soil series (Humic Udivitrant) is the youngest soil series, still showing clear andic properties. This soil type is characterized by a brownish black (7.5YR 2/1) A horizon, with a loamy texture and a moderate, very fine to fine subangular blocky structure, gradually grading into a dull, yellowish brown (10YR 5/6) cambic B horizon with a loamy sand texture and a weak, fine massive structure, on a dull yellowish brown (10YR 5/6) C horizon with a sand texture.

Suitable research sites were selected using georeferenced descriptions based on augerings and representative profile descriptions, all stored in a geographic information system. To avoid differences in soil physical properties due to different management histories, only sites that had been in banana plantations for >8 yr were selected. On carefully selected locations for each soil series, soil profile descriptions were made. Samples were collected for texture analysis and determination of organic matter content and soil moisture content in the high pressure head range. In the two upper horizons of each soil, five undisturbed core samples, with diameter and height of 0.07 m, were sampled for hydraulic parameter estimation with the one-step outflow procedure. After conducting the tests, 27 of the 30 samples were accepted for statistical analysis. Also field measurements with the modified crust test were made (Booltink et al., 1991). On each site, six soil columns with a height and diameter of 0.30 m were installed to obtain data on saturated and unsaturated hydraulic conductivity in the pressure head range from 0 to -3.0 kPa. Three of the columns were installed in the topsoil, to characterize the 0–0.30-m layer, while two columns were installed in the subsoil, to characterize the 0.30–0.60-m layer. One separate measurement was made in the topsoil, to characterize the saturated hydraulic conductivity (K_{sat}) without the soil being disturbed

by applying and removing the crust. In total, 11 measurements were completed in the topsoil and five in the subsoil.

Characterizing Soil Physical Properties

Laboratory Measurements

Continuous equations describing soil hydraulic parameters were obtained using cumulative outflow data, measured as a function of time, as input in the SFIT parameter estimation procedure. Cumulative outflow data were obtained with the one-step outflow method (Doering, 1965), applying a step change in air pressure on top of the undisturbed soil cylinders (Kool et al., 1985; Parker et al., 1985). Only a short outline is presented here.

Soil hydraulic parameters were described with the analytical parameterized equations of van Genuchten (1980), according to (Wösten and van Genuchten, 1988)

$$\theta(h) = \theta_r + \frac{(\theta_s - \theta_r)}{[1 + (\alpha h)^n]^m} \quad [1]$$

and

$$K(h) = K_{sat} \frac{(\{[1 + (\alpha h)^n]^m\} - (\alpha h)^{n-1})^2}{[1 + (\alpha h)^n]^{m(\gamma+2)}} \quad [2]$$

where θ_r is the residual water content, θ_s the saturated water content, h is the pressure head (kPa), K_{sat} is the saturated hydraulic conductivity (m s^{-1}) eliminating bypass flow (Bouma, 1982). Ponding of water on detached soil cores results in much higher K_{sat} values that are nonrepresentative for natural conditions (Bouma, 1991), α approximates the reciprocal of the air-entry value (cm^{-1}), and n , m , and γ are dimensionless fitting parameters. Parameters in the equations were optimized by comparing simulated and measured cumulative outflow as a function of time, using an optimization algorithm based on Marquardt's maximum neighborhood method. An objective function has to be minimized, according to

$$E(b) = \sum_{i=1}^n \{w_i [Q(t_i) - \hat{Q}(t_i, b)]\}^2 + \sum_{j=1}^m \{v_j [\theta(h_j) - \hat{\theta}(h_j, b)]\}^2 \quad [3]$$

in which w_i and v_j are weights assigned to measured (Q) and predicted outflow ($\hat{Q}$) or moisture content (θ and $\hat{\theta}$), respectively (Parker et al., 1985).

Measured $\theta(h)$ points were also used in the objective function, because previous experiments showed that with the combined use of outflow and $\theta(h)$, a reduction in prediction error was possible. The parameters θ_s and K_{sat} were not optimized but obtained with independent experiments. Initial values for the parameter θ_r were set at the measured -1.5 MPa. Initial values for α , n , and γ were obtained from data sets available for selected textural classes in Dutch soils (Wösten et al., 1987). Fixed basic input data for θ_s were obtained from independent saturation experiments using five undisturbed $0.1 \times 10^{-3} \text{ m}^3$ cores per horizon that were slowly saturated.

Field Measurements

Input data for K_s were obtained with the modified crust test, measuring the water flow at zero pressure head using a crust to eliminate bypass flow. Using input data from measurements at zero pressure applying a crust was considered necessary because the pore size distribution model of Mualem, included in the equations of van Genuchten, does not take bypass flow

into account (Booltink et al., 1991). Detailed information on the theory and use of the modified crust method can be found in Bootink et al. (1991).

Simulation Modeling

To study the effect of soil physical parameters on the functional properties (water supply, crop yield, and water flux below the root zone), use was made of the simulation model SWACROP. SWACROP is the result of integrating SWATRE (Soil Water Actual Transpiration Rate—Extended), which describes water flow in a heterogeneous soil-root system (Belmans et al., 1983), with CROPR (CROp PRoduction), which calculates crop dry matter production in accordance with soil water pressure head (Feddes et al., 1988). The calculation of the one-dimensional water balance is based on a transient finite-difference soil-water model. Water uptake by roots is represented as a sink that depends on the soil water pressure head. The basic equation for water flow is

$$\frac{\partial h}{\partial t} = \frac{1}{C(h)} \frac{\partial}{\partial z} \left[K(h) \left(\frac{\partial h}{\partial z} + 1 \right) \right] - \frac{S(h)}{C(h)} \quad [4]$$

where h = soil water pressure head (kPa, negative in unsaturated soil), t = time (s), C = differential moisture capacity $d(\theta)/dh$, θ being the volumetric soil water content ($\text{m}^3 \text{ m}^{-3}$), z = vertical distance (m) with the origin at the soil surface and directed positive upward, K = hydraulic conductivity (m s^{-1}), and S = water uptake by roots (s^{-1}). The value of S is calculated according to the procedure described by Feddes et al. (1978) and Prasad (1978):

$$S(h) = \alpha(h) S_{max} \quad [5]$$

where $\alpha(h)$ is a dimensionless function of pressure head and S_{max} is the maximum possible water extraction by roots:

$$S_{max}(z) = \frac{2T_p}{z_r} \left(1 - \frac{z}{z_r} \right) \quad [6]$$

where T_p is the potential transpiration, and z_r is the depth (m) of the root zone, and z is the depth (m) below the soil surface.

The α function is presented in Fig. 1. Water uptake below h_0 is zero due to anaerobiosis, between h_0 and h_1 α linearly increases, and it is set to unity between h_1 and h_2 . Between h_2 and h_3 , α decreases linearly and becomes zero at h_3 (wilting point). The value of h_2 is dependent on the evaporative demand and so varies with T_p .

Simulated actual water extraction by roots ($\text{m}^3 \text{ m}^{-3} \text{ s}^{-1}$) can be transformed to actual transpiration (m s^{-1}), which can be used to calculate the actual dry matter production q_a ($\text{kg m}^{-2} \text{ s}^{-1}$), according to

$$q_a = 0.5 \left[A \frac{T}{\Delta < e >} + q_m - \sqrt{\left(q_m + A \frac{T}{\Delta < e >} \right)^2 - 4 q_m A \frac{T}{\Delta < e >} (1 - \xi)} \right] \quad [7]$$

where A = maximum water use efficiency ($\text{kg m}^{-3} \text{ kPa}$), T = actual transpiration rate (m s^{-1}), $\Delta < e >$ = average vapor pressure deficit of the air (kPa), q_m = maximum possible growth rate ($\text{kg m}^{-2} \text{ s}^{-1}$), and ξ = mathematical parameter (0.01).

Using daily climatic data on precipitation (m d^{-1}), net radiation (W m^{-2}), temperature ($^{\circ}\text{C}$), humidity (%), and sun hours,

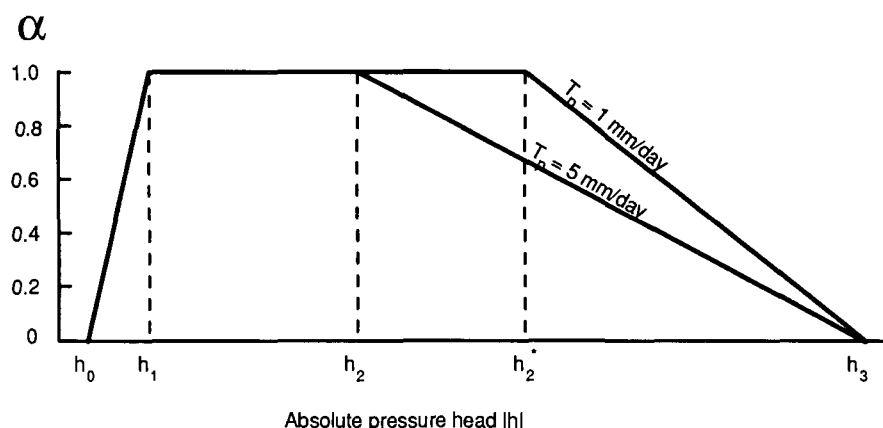


Fig. 1. Dimensionless sink term variable α as a function of the absolute value of the soil water pressure head.

the potential transpiration (m d^{-1}) with the corresponding potential dry matter production can also be calculated (Smith, 1991). Climatic data were obtained from the El Carmen weather station in the Atlantic Zone of Costa Rica, using the wet growing season of 1988, the average growing season of 1991, and the dry growing season of 1985. Simulated crop production and percolation fluxes were calculated for the crop maize, grown in the months January to May. Some basic climatic data are shown in Table 2.

Measured soil physical parameters were incorporated in different forms into the simulation model. First, soil water retention and hydraulic conductivity were expressed as continuous functions, using the parameters of van Genuchten (1980), determined with the SFIT parameter estimation procedure. Second, water retention and hydraulic conductivity were expressed in tabular form, using discrete steps of 0.01 m m^{-3} soil water content with related pressure head and hydraulic conductivity. The second approach offered the possibility of combining results of different methods (crust test and parameter estimation) to obtain data on soil physical parameters. In this approach modified crust test data were used in the pressure head range of 0 to -2.0 kPa , and the parameterized equations of van Genuchten were employed for the lower pressure heads.

To run the simulation model, characterized soil layers were combined into soil profiles. Because data obtained with the modified crust test only covered the upper 0.6 m, soil profiles used in SWACROP were limited to this depth. To take the observed variation in physical properties into account, all possible combinations of top- and subsoils were used in SWACROP. These different combinations of top- and subsoils lead to approximately 25 to 30 different simulations runs for each soil and year (see Tables 3 and 4).

The simulation model was only used to express the effects of different hydraulic characteristics on two calculated functional properties; all other parameters and boundary conditions in the model were identical in all calculations. Thus, the main objective was a sensitivity analysis, not predicting yields or

fluxes. The sensitivity analysis, as presented, is unusual as it is restricted to only ranges of measured values rather than to hypothetical ranges. Field monitoring for validation purposes was not part of this study. Good results obtained with the SWACROP model at other locations (e.g., van Lanen et al., 1986; Feddes et al., 1988; de Jong and Kabat, 1990; Ragab et al., 1990a,b; Zepp and Belz, 1992) provide adequate justification for its use in a sensitivity analysis.

RESULTS AND DISCUSSION

Physical Characteristics

The results of the SFIT parameter estimation procedure are shown in Fig. 2 through 4 for Ligia, Montelimar, and Los Diamantes soils, respectively. The curves obtained for the different replicate samples in any given soil layer are quite similar. Figures 2 through 4 also illustrate the difficulties of fitting the conductivity curve with the independently measured saturated and unsaturated hydraulic conductivities in the pressure head range 0 to -3.0 kPa . The inadequacy is caused by the high macroporosity of the soil layers under study; as mentioned above, this property differs from conditions used to develop the pore size distribution model of Mualem, which is incorporated in the closed-form equations of van Genuchten. Thus, extrapolation of measured outflow data beyond the -5.0 to -60.0 kPa pressure step used to optimize the van Genuchten equations is not realistic, particularly at high pressure heads.

Effect of Different Types of Physical Input Data

The averaged results for the functional properties crop yield and percolation flux are shown in Tables 3 and 4 for three soil types, three growing seasons, and two types of basic input. To see whether a different approach in handling input data resulted in significantly different results in crop yield and percolation fluxes, variance analyses were made.

A standard completely randomized variance analysis of the simulation results of all possible combinations of top- and subsoils shows that the tabular input of soil physical data into the simulation model results in significantly (5%) higher yields than when only the continuous equations described with the van Genuchten parameters

Table 2. Basic climate data from the El Carmen weather station in the Atlantic Zone of Costa Rica.

	1985	1988	1991
Yearly precipitation, mm	2810.1	3602.2	4077.4
Precipitation during growing season, mm	526.1	1393.6	939.5
Average temperature during growing season, °C	24.7	26.1	26.1
Average relative humidity during growing season, %	81	80	80
Average radiation during growing season, $\text{W m}^{-2} \text{ d}^{-1}$	295.8	268.3	267.6

Table 3. Yield results of simulation crop production with SWACROP for three growing seasons, three soil series, and two approaches for handling basic physical input data.

	Los Diamantes			Ligia			Montelimar			Average yield
	1985	1988	1991	1985	1988	1991	1985	1988	1991	
	kg m ⁻²									
Tabular input										
Yield, kg m ⁻²	0.5518	0.5603	0.4667	0.3485	0.4713	0.3180	0.5993	0.5596	0.4664	0.4835
Standard deviation	0.0073	0.0039	0.0076	0.0978	0.0667	0.664	0.0090	0.362	0.0529	
No. of simulations	25	25	25	36	36	33	30	30	30	
Continuous input										
Yield, kg m ⁻²	0.5630	0.4389	0.3020	0.3129	0.3248	0.2158	0.5999	0.5630	0.4369	0.4175
Standard deviation	0.0189	0.0953	0.0774	0.1010	0.1027	0.0539	121	48	588	
No. of simulations	25	22	25	22	20	25	30	30	30	
Average yield, kg m ⁻²										
Tabular input		0.5263			0.3793			0.5418		
Continuous input		0.4171			0.2845			0.5333		

are used. An example of differences in simulated daily soil water pressure heads during the growing season is shown in Fig. 5. The figure shows that using only the continuous van Genuchten function results in the simulation of wetter soil conditions throughout the growing season. Also, the pressure head falls more frequently above the level of -3.0 hPa, the level (h_0) at which O₂ shortage, associated with air contents lower than 0.01 m m⁻³, is assumed to begin. Wetter soil conditions are caused by the lower estimated unsaturated hydraulic conductivities in the pressure head range near saturation, resulting from ignoring the presence of macroporosity. The level of -10.0 kPa (h_i), the level at which O₂ stress becomes nonlimiting, is also included in Fig. 5.

Significant differences (at the 0.05 level) between the two approaches for handling input data can also be observed when considering the simulated percolation fluxes (Table 4). The amount of percolating water is higher when use is made of one continuous function instead of a table containing the combined results of different techniques. The differences, however, are not very striking, which is probably the result of counterbalancing processes. Including macroporosity in basic input data for simulation results in quicker drainage of water to the subsoil after a rain shower, which results in drier soil conditions and more favorable growing conditions. This results in more crop water uptake and transpiration, so that the total amount of percolation losses is smaller.

This study did not include actual field monitoring of pressure heads, which should be part of future studies

to allow selection of the most appropriate method. However, measurements elsewhere in soils with macropores have clearly demonstrated that ignoring the presence of macropores may yield poor prediction of pressure head (e.g., White, 1985; Booltink and Bouma, 1991). We, therefore, conclude not only that the two procedures for expressing hydraulic characteristics are different but that the tabular input data is probably most realistic.

Functional Differences between Soil Types

To decide on the allowable level of "pooling" of soil types, so as to reduce as much as possible the amount of input data necessary for simulation modeling, simulation results based on tabular input data are used (Tables 3 and 4). Again, crop yield and percolation fluxes are considered as indicators.

Variance analyses show that there is a significant difference in maize yield between the Los Diamantes and Montelimar soils on the one hand and the Ligia soil on the other (Table 3). The subdivision into the two mentioned groups corresponds with differences in texture. Los Diamantes and Montelimar are more sandy, while Ligia is more clayey. Crops on clayey soil types suffer more from prolonged wetness, while in sandy soils more rapid water percolation of water leads to a faster recovery of optimal growing conditions. These results indicate that, on the basis of simulated crop yields,

Table 4. Results for percolation fluxes simulated with SWACROP for three growing seasons, three soil series, and two approaches for handling basic physical input data.

	Los Diamantes			Ligia			Montelimar			Average flux
	1985	1988	1991	1985	1988	1991	1985	1988	1991	
	m season ⁻¹									
Tabular input										
Flux, m season ⁻¹	-0.18	-0.96	-0.49	-0.19	-0.95	-0.56	-0.18	-0.96	-0.53	-0.55
Standard deviation	0.002	0.003	0.020	0.081	0.331	0.034	0.007	0.009	0.035	
No. of simulations	25	25	25	36	36	33	30	30	30	
Continuous input										
Flux, m season ⁻¹	-0.19	-0.99	-0.54	-0.25	-0.99	-0.60	-0.17	-0.96	-0.55	-0.58
Standard deviation	0.010	0.056	0.036	0.041	0.0146	0.053	0.010	0.001	0.044	
No. of simulations	25	22	25	22	20	25	30	30	28	
Average flux										
Tabular input		-0.54			-0.57			-0.55		
Continuous input		-0.58			-0.61			-0.56		

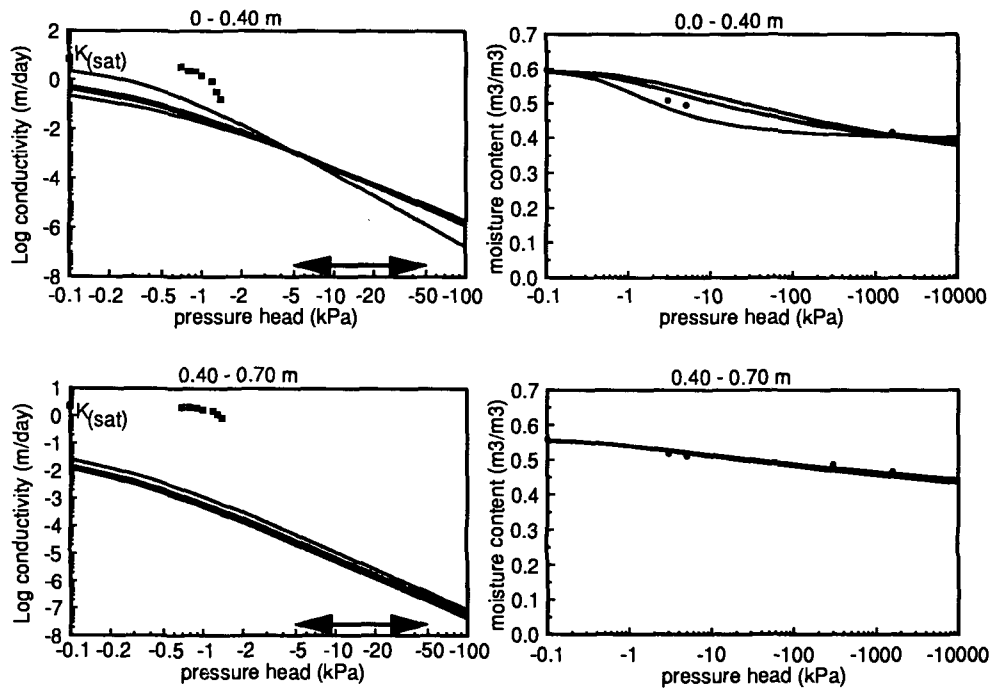


Fig. 2. Water retention and hydraulic conductivity curves for the soil series Ligia, as obtained with the parameter estimation procedure for the closed-form equations of van Genuchten (1980). Independent measurement results of the modified crust method are shown as well (solid squares). The arrow indicates the range of pressure heads directly associated with the one-step outflow method.

the three soil series should be combined into not one but two functional groups.

No significant differences in percolation fluxes in the different soil types are found when use is made of a tabular

input for soil hydraulic parameters (Table 4). This absence of differences is most likely caused by counterbalancing processes, a more rapid drainage combined with better growing conditions and water uptake by crops on sandy

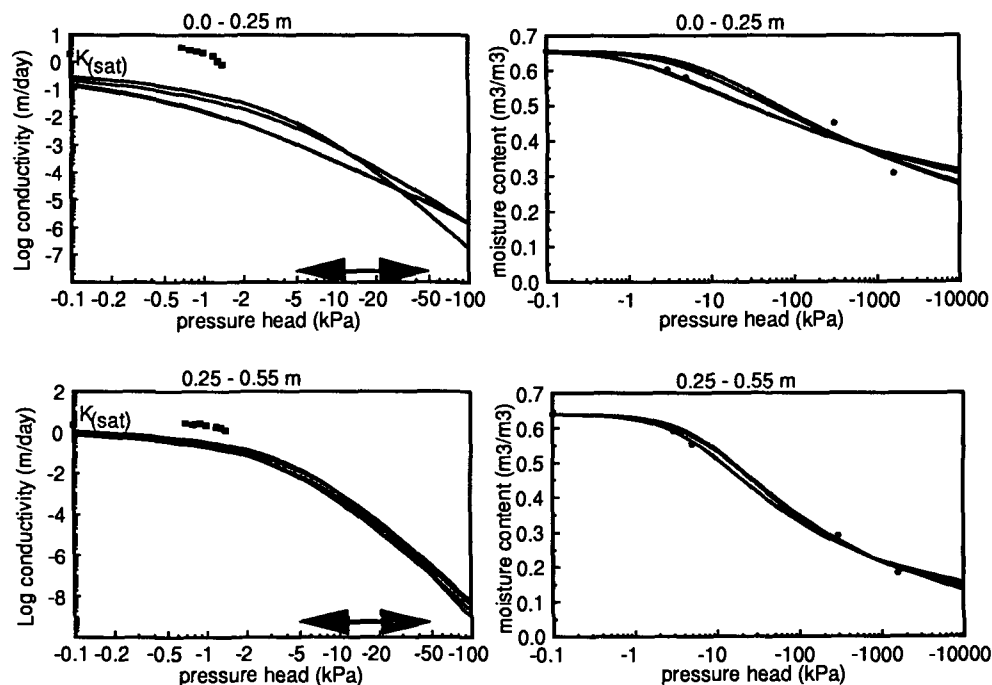


Fig. 3. Water retention and hydraulic conductivity curve for the soil series Montelimar, as obtained with the parameter estimation procedure for the closed-form equations of van Genuchten (1980). Independent measurement results of the modified crust method are shown as well (solid squares). The arrow indicates the range of pressure heads directly associated with the one-step outflow method.

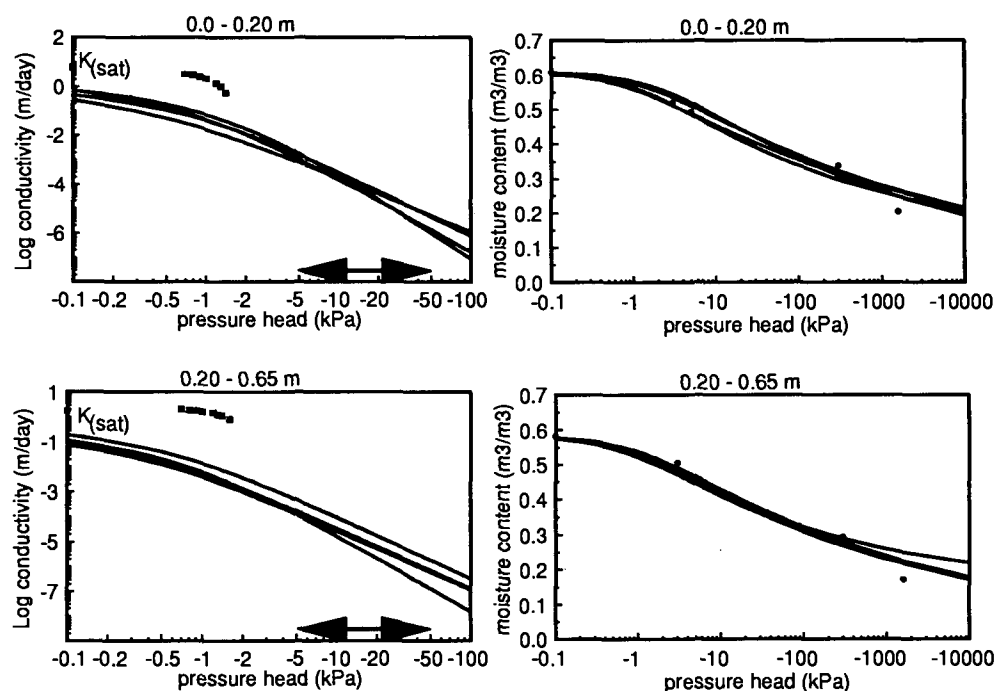


Fig. 4. Water retention and hydraulic conductivity curve for the soil series Los Diamantes, as obtained with the parameter estimation procedure for the closed-form equations of van Genuchten (1980). Independent measurement results of the modified crust method are shown as well (solid squares). The arrow indicates the range of pressure heads directly associated with the one-step outflow method.

soil series vs. slower drainage combined with more unfavorable growing conditions in the clayey soil series. Thus, based on the criterion of percolation flux, soils could be combined into one functional group.

CONCLUSIONS

Hydraulic conductivity and moisture retention data are needed for simulating water fluxes and crop growth.

Measurements with the one-step outflow method in the pressure head range between -5.0 and -60.0 kPa, and expressed in terms of the van Genuchten parameters, are often extrapolated to the pressure range between 0 and -5.0 kPa. However, hydraulic conductivity values thus obtained in three Costa Rican soil series differed significantly from values measured by the modified crust test procedure in the pressure range between 0 and -2.0

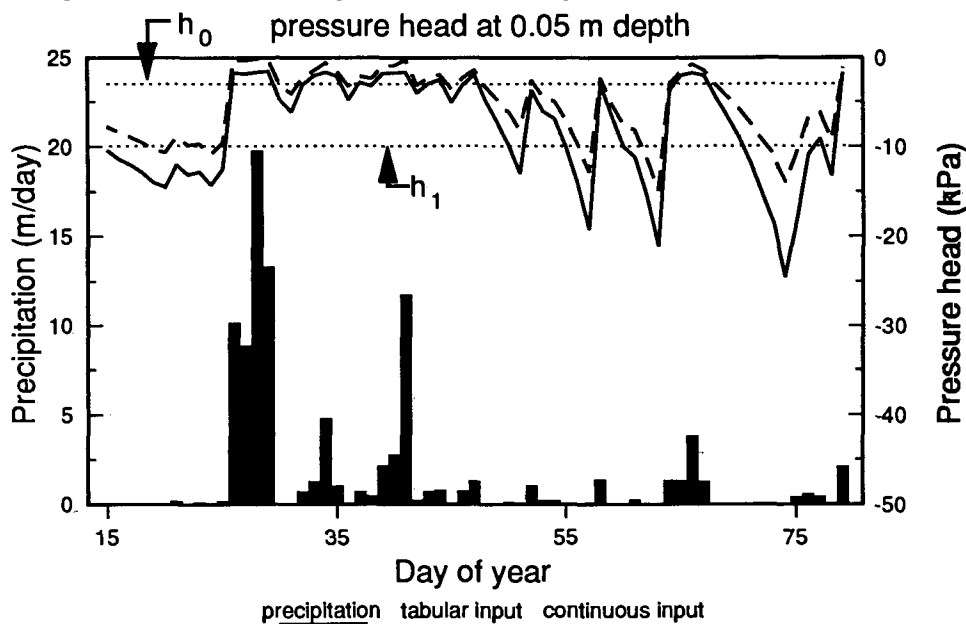


Fig. 5. Example of the simulated daily fluctuations of soil water pressure on Los Diamantes soil during the growing season of 1988, comparing two approaches for handling basic physical input data. The dotted lines indicated as h_0 and h_1 represent the levels of complete anaerobiosis and nonstressed water uptake according to Fig. 1, respectively.

kPa. Corresponding simulated maize yields and fluxes below the root zone were also significantly different. Field testing is needed to further evaluate errors arising from extrapolation of widely used one-step outflow and instantaneous-profile data to pressure heads near saturation in soils with macropores.

Using tabular hydraulic data derived from the modified crust test and the one-step outflow method, maize yields and fluxes below the root zone were calculated for 3 yr and three Costa Rican soil series belonging to one broad category of young, well-drained, fertile Holocene soils. Comparing yields and fluxes served as a functional analysis and showed that a distinction of two categories would have been appropriate when considering maize yields, while distinction of the one broad category was justified when considering fluxes below the root zone. The example presents a methodology to combine different soil series into functionally identical groups, which is desirable to reduce data needs and computer time in interdisciplinary projects.

ACKNOWLEDGMENTS

The area of study was part of the Agricultural Research Program in the Atlantic Zone of Costa Rica. The program is carried out in cooperation between the Centro Agronomico Tropical de Investigacion y Enseñanza (CATIE), the Ministerio de Agricultura y de Ganaderia (MAG), and Wageningen Agricultural University. The central theme of the research program is to develop a methodology for analysis and planning of ecologically and economically sustainable land use. The authors acknowledge assistance by Ir. M.J.D. Hack-ten Broeke of the SC-DLO for using the SWACROP model and Mrs. A.M. Weitz of the Atlantic Zone Program for assistance in making physical measurements.

REFERENCES

- Alfaro, R., J. Bouma, L.O. Fresco, D.M. Janssen, S.B. Kroonenberg, A.C.J. van Leeuwen, R.A. Schipper, R.J. Sevenhuysen, J.J. Stoorvogel and V. Watsin. 1994. Sustainable land use planning in Costa Rica. In L.O. Fresco et al. (ed.) *The future of the land*. Kluwer Publ. Co., Dordrecht, the Netherlands (in press).
- Belmans, C., J.G. Wesseling, and R.A. Feddes. 1983. Simulation model of the water balance of a cropped soil: SWATRE. *J. Hydrol.* (Amsterdam) 63:271-286.
- Booltink, H.W.G., and J. Bouma. 1991. Physical and morphological characterization of bypass flow in a well-structured clay soil. *Soil Sci. Soc. Am. J.* 55:1249-1254.
- Booltink, H.W.G., J. Bouma, and D. Gimenez. 1991. Suction crust infiltrometer for measuring hydraulic conductivity of unsaturated soils near saturation. *Soil Sci. Soc. Am. J.* 55:566-568.
- Bouma, J. 1982. Measuring the hydraulic conductivity of soil horizons with continuous macropores. *Soil Sci. Soc. Am. J.* 46:438-441.
- Bouma, J. 1991. Influence of soil macroporosity on environmental quality. *Adv. Agron.* 46:1-37.
- Breeuwsma, A., J.H.M. Wösten, J.J. Vleeshouwer, A.M. van Slobbe, and J. Bouma. 1986. Derivation of land qualities to assess environmental problems from soil surveys. *Soil Sci. Soc. Am. J.* 50:186-190.
- de Jong, R., and P. Kabat. 1990. Modelling water balance and grass production. *Soil Sci. Soc. Am. J.* 54:1725-1732.
- Doering, E.J. 1965. Soil-water diffusivity by the one-step method. *Soil Sci.* 99:332-326.
- Feddes, R.A., P. Kabat, P.J.T. van Bakel, J.J.B. Bronswijk, and J. Halbertsma. 1988. Modelling soil water dynamics in the unsaturated zone—State of the art. *J. Hydrol.* (Amsterdam) 100:69-111.
- Feddes, R.A., P.J. Kowalik, and H. Zaradny. 1978. Simulation of field water use and crop yield. *Simul. Monogr. PUDOC*, Wageningen, the Netherlands.
- Kool, J.B., J.C. Parker, and M.Th. van Genuchten. 1985. Determining soil hydraulic properties from one-step outflow experiments by parameter estimation: I. Theory and numerical studies. *Soil Sci. Soc. Am. J.* 49:1348-1354.
- Parker, J.C., J.B. Kool, and M.Th. van Genuchten. 1985. Determining soil hydraulic properties from one-step outflow experiments by parameter estimation: II. Experimental studies. *Soil Sci. Soc. Am. J.* 49:1354-1359.
- Prasad, R. 1988. A linear root water uptake model. *J. Hydrol.* (Amsterdam) 99:297-306.
- Ragab, R., F. Beese, and W.W. Ehlers. 1990a. A soil water balance and dry matter production model: I. Soil water balance of oat. *Agron. J.* 82:152-156.
- Ragab, R., F. Beese, and W.W. Ehlers. 1990b. A soil water balance and dry matter production model: II. Dry matter production of oat. *Agron. J.* 82:157-161.
- Smith, M. 1991. Report on the expert consultation on procedures for revision of FAO guidelines for prediction of crop water requirements, 28-31 May 1990, Rome. Land and Water Dev. Div., FAO, Rome.
- van Genuchten, M.Th. 1980. A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Sci. Soc. Am. J.* 44:892-898.
- van Lanen, H.A.J., M.H. Bannink, and J. Bouma. 1986. Use of simulation to assess the effect of different tillage practices on land qualities of a sandy loam soil. *Soil Tillage Res.* 10:347-361.
- White, R.E. 1985. The influence of macropores on the transport of dissolved and suspended matter through the soil. *Adv. Soil Sci.* 3:95-121.
- Wielemaker, W.G., and S.B. Kroonenberg. 1992. Generacion y aplicacion de la informacion de suelos de la Zona Atlantica de Costa Rica, actas del taller informacion de suelos, Guápiles, Costa Rica. 2-4 Octubre 1990. Programa Zona Atlantica, CATIE-UAW-MAG, Turrialba, Costa Rica.
- Wösten, J.H.M., M.H. Bannink, and J. Beuver. 1987. Water retention and hydraulic conductivity characteristics of top- and subsoils in the Netherlands: The STARING series. (In Dutch.) ICW Rep. 18. Instituut voor Cultuurtechniek en Waterhuishouding, Wageningen, the Netherlands.
- Wösten, J.H.M., J. Bouma, and G.H. Stoffelsen. 1985. The use of soil survey data for regional soil water simulation models. *Soil Sci. Soc. Am. J.* 49:1238-1245.
- Wösten, J.H.M., and M.Th. van Genuchten. 1988. Using texture and other soil properties to predict the unsaturated soil hydraulic functions. *Soil Sci. Soc. Am. J.* 52:1762-1770.
- Zepp, H., and A. Belz. 1992. Sensitivity and problems in modelling soil moisture conditions. *J. Hydrol.* (Amsterdam) 131:227-238.